Duncan Larzelere

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Education

University of Michigan, College of LSA, Honors Program Ann Arbor, MI
Bachelor of Science in Honors Mathematics and Computer Science Expected May 2029

- Sidney J. and Irene Shipman Scholar: $20,000 per year merit award, about 16 recipients per entering class. SAT: 1600.
- Coursework (Fall 2026): Putnam Problem Solving Seminar (Math 289), Honors Mathematics I (Math 295), Introduction to Combinatorics and Graph Theory (Math 465), Programming and Introductory Data Structures (EECS 280).

Michigan State University, Dual Enrollment East Lansing, MI
University mathematics coursework, GPA: 4.0 2024 - 2026

- Coursework: Honors Real Analysis (measure theory, point-set topology, Baire category theorem), Linear Algebra (proof-based), Multivariable Calculus, Differential Equations.

Work Experience

Freelance Full-Stack Developer, Self-employed, Remote 2023 - 2025

- Built a storefront for composer John Danziger (johndanziger.com) that takes card payments, delivers files automatically on purchase, and lets the client upload new work from an admin page, using Next.js, Stripe, and Supabase.
- Built PupPrints, a pet-portrait service that fine-tunes an image model per dog using Replicate, handling upload, payment, and delivery; processed 2,000+ images.

Founder and Game Developer, DL8 Studios, Remote 2020 - 2023

- Built and published Drifty Car, an iOS racing game with custom physics and procedural tracks (Unity, C#, Firebase); 17,000+ views across 43 countries.

Projects

Melody Diffuser (PyTorch, CUDA, Diffusion Transformers), demo at dl8ai.com 2025

- Built a 32M-parameter discrete diffusion transformer (6 layers, 8 heads, RMSNorm, SwiGLU, weight-tied embeddings) that writes four-bar melodies in 64 denoising steps instead of note by note.
- Extracted and tokenized 10M four-bar melodies from raw MIDI for training, as no public dataset was large enough.
- Added gesture control at inference using MediaPipe hand tracking, training with classifier-free guidance so the model runs with or without gesture input; labeled gesture-melody pairs with a rule-based pipeline over note jumps.
- Fixed a training plateau caused by token class imbalance by reweighting the loss against spurious hold predictions.

Otto, Adaptive Poker AI (TypeScript, Monte Carlo CFR, Reinforcement Learning), demo at dl8ai.com 2026

- Built a heads-up no-limit hold'em AI from scratch: an equilibrium strategy from 66M iterations of Monte Carlo CFR self-play, tilted by a PPO-trained policy to exploit habits read from 16 Bayesian posteriors.
- Won 2 to 7 times more than equilibrium play against every leaky style, and more against 39 of 40 unseen opponents.

Prediction-Market Forecasting Engine (TypeScript, Next.js, Monte Carlo simulation) 2026

- Built a live win-probability engine for a fantasy World Cup draft from 54 Kalshi markets, updating as games finish.
- Simulated 50,000 full brackets per update so elimination dependencies between teams are preserved.
- Dropped closed, illiquid, and stale markets from the pricing feed.

Motorized Ping-Pong Launcher (Python, Raspberry Pi, Mechatronics) 2026

- Built a joystick-aimed ball launcher with dual-flywheel firing, servo aiming, and a sensor that clears feeder jams.

Awards

- Kaggle Notebooks Silver Medal: 9th of 955 community notebooks, March Madness tournament prediction.
- Hack with the Beat: Best Machine Learning Project, 2nd Best Individual Overall, 3rd Best Presentation.

Skills

- Programming Languages: Python, C++, TypeScript, C#
- Frameworks and Tools: PyTorch, Transformers, NumPy, XGBoost, CUDA, Git, Next.js, Supabase, Stripe, Unity
- Mathematics: Real Analysis, Probability, Linear Algebra, Combinatorics
- Interests: Poker, Chess, Water Skiing, Classical Music